



Palo Alto Networks Cybersecurity Academy – Setting up a PA220 Firewall for a SOHO Environment

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Purpose

This lab provides the necessary knowledge to secure a Small Office Home Office or SOHO network using a Palo Alto 220 Firewall. The configuration of the SOHO network using a Palo Alto firewall gives familiarity with other Palo Alto devices and gives knowledge on the general security guidelines for configuring other SOHO networks. The lab additionally provides experience on how firewalls function to protect networks.

Background

A Small Office Home Office or SOHO network is a network consisting of around one to ten people. These networks can be found in homes and small businesses with few devices. SOHO networks differ greatly from enterprise networks not just in size but also in terms of complexity. In comparison to Enterprise networks SOHO networks are significantly simpler typically consisting typically of only one firewall, router, and switch that connects directly to end devices like computers and printers granting access to the internet. The Firewall in these SOHO networks needs to effectively serve as the gateway to the Internet Service Provider or ISP to ensure that the small business or home can reach the internet.

The Firewall is considered one of the most important pieces of hardware in a SOHO network, if not then the most important part of a SOHO network as it serves as the strongest most reliable line of defense between attackers and small businesses. Small businesses face constant cyber threats with statistics showing that 40-72% of small businesses report that they face cyber threats, with some reporting that they face cyber-attacks every 11 seconds. These attacks are directed at small businesses who are seen to have weaker network infrastructure and security, making an ideal target for hackers to exploit them. Due to these constant threats the importance of firewalls in SOHO networks is amplified creating the need for network administrators to configure firewalls to filter, monitor, and flag suspicious traffic so that these small businesses can keep functioning. The most common cyber-attacks against small businesses are DDoS attacks, Malware, and Phishing attacks. The attack that most commonly affects small businesses is the DDoS attacks that exploit old software or code that is not consistently updated against newer and more dangerous attacks. After getting access the exploiters create botnets that flood the network preventing access to the internet, possibly leading to that business shutting down.

For a Firewall to keep a business safe first must be placed at a point where all network traffic from the internet and the SOHO network are required to travel through the firewall this usually means that the firewall is the last device sitting between the ISP and the network. Firewalls keep businesses safe from DDoS attacks as long as the firewall is consistently being updated to fight against the newest cyber threats

especially as advancements in AI and quantum computing continue to be utilized for finding holes in software and exploiting them to gain access to user and customer accounts and information. These updates are sent out constantly by the company who designed the firewall, in the case of this lab: Palo Alto, but require configuration to automatically download and install once received.

Lab Summary

In this lab you configure a Palo Alto 220 Firewall for a SOHO network. The firewall has one interface connecting to the ISP which the ISP will dynamically assign an IP address for that interface. The firewall will also use DHCP to dynamically assign IP addresses for the end devices in the SOHO network, perform PAT a type of network address translation to protect internal IP address, configure trusted and untrusted ports.

Lab Commands

Create Network Zones

Go to Network > Zones > Add

Create 3 Zones:

Untrust-L3:

The screenshot shows the 'Zone' configuration page in the Palo Alto Networks management console. The 'Name' field is set to 'Untrust-L3'. The 'Log Setting' is 'None' and the 'Type' is 'Layer3'. Under the 'INTERFACES' section, there is an 'Add' button. The 'Zone Protection' section shows 'Zone Protection Profile' set to 'None' and 'Enable Packet Buffer Protection' checked. The 'User Identification ACL' section has 'Enable User Identification' unchecked, an 'INCLUDE LIST' with an 'Add' button, and an 'EXCLUDE LIST' with an 'Add' button. The 'Device-ID ACL' section has 'Enable Device Identification' unchecked, an 'INCLUDE LIST' with an 'Add' button, and an 'EXCLUDE LIST' with an 'Add' button. At the bottom right, there are 'OK' and 'Cancel' buttons.

Trust-L3:

Zone

?

Name

Trust-L3

Log Setting

None

Type

Layer3

INTERFACES

+

Add

-

Delete

Zone Protection

Zone Protection Profile

None

Enable Packet Buffer Protection

☒

User Identification ACL

Enable User Identification

☐

INCLUDE LIST

+

Add

-

Delete

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

Users from these addresses/subnets will be identified.

EXCLUDE LIST

+

Add

-

Delete

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

Users from these addresses/subnets will not be identified.

Device-ID ACL

Enable Device Identification

☐

INCLUDE LIST

+

Add

-

Delete

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

Devices from these addresses/subnets will be identified.

EXCLUDE LIST

+

Add

-

Delete

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

Devices from these addresses/subnets will not be identified.

OK

Cancel

Trust-L2:

Zone

?

Name

Trust-L2

Log Setting

None

Type

Layer2

INTERFACES

+

Add

-

Delete

Zone Protection

Zone Protection Profile

None

Enable Packet Buffer Protection

☒

User Identification ACL

Enable User Identification

☐

INCLUDE LIST

+

Add

-

Delete

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

Users from these addresses/subnets will be identified.

EXCLUDE LIST

+

Add

-

Delete

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

Users from these addresses/subnets will not be identified.

Device-ID ACL

Enable Device Identification

☐

INCLUDE LIST

+

Add

-

Delete

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

Devices from these addresses/subnets will be identified.

EXCLUDE LIST

+

Add

-

Delete

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

Devices from these addresses/subnets will not be identified.

OK

Cancel

This configuration should result to the following:

PA-220

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Portals

Gateways

MDM

Connect the Firewall to the ISP

Connect an Ethernet Cable to from the Firewall's ethernet1/1 port to the switch connected to the ISP

Go to Network > Interfaces > ethernet1/1

Configure it as Layer 3, Default Virtual Router, Untrust-L3

Ethernet Interface

Interface Name

ethernet1/1

Comment

Interface Type

Layer3

Netflow Profile

None

Config

IPv4

IPv6

SD-WAN

Advanced

Assign Interface To

Virtual Router

default

Security Zone

Untrust-L3

OK

Cancel

Navigate to IPV4 and set the interface as a DHCP Client and check “Enable” and “Automatically create a default route pointing to default gateway provided by server”. This port is configured as untrusted due to it being the port that will receive packets from the internet from untrusted or unknown sources.

Ethernet Interface ?

Interface Name

ethernet1/1

Comment

Interface Type

Layer3

Netflow Profile

None

Config

IPv4

IPv6

SD-WAN

Advanced

☐ Enable SD-WAN
 ☐ Enable Bonjour Reflector

Type
 ☐ Static
 ☐ PPPoE
 ☒ DHCP Client

☒ Enable

☒ Automatically create default route pointing to default gateway provided by server

☐ Send Hostname

system-hostname

Default Route Metric

10

[Show DHCP Client Runtime Info](#)

OK

Cancel

Create a Vlan Object

Go to Vlan > Network > and click Add

Create a name and set the interface to vlan

VLAN ?

Name

Vlan Object

VLAN Interface

vlan

☐ INTERFACES ^

MAC ADDRESS	INTERFACE

+ Add

- Delete

Static MAC Configuration

MAC ADDRESS	INTERFACE

+ Add

- Delete

OK

Cancel

Configure the Layer 2 Ports and Vlan Object

Go to Network > Interface > Ethernet

For Ethernet1/2 and Ethernet1/3 Configure:

Interface Type: Layer 2

Netflow Profile: None

VLAN: Vlan Object

Security Zone: Trust-L2

Example of Ethernet1/2:

Ethernet Interface

Interface Name

ethernet1/2

Comment

Interface Type

Layer2

Netflow Profile

None

Config

Advanced

Assign Interface To

VLAN

Vlan Object

Security Zone

Trust-L2

OK

Cancel

Configure the Vlan Interface

Go to Network > Interfaces > VLAN

Set VLAN to Vlan Object, Virtual Router to Default, Security Zone to Trust-L3

VLAN Interface

Interface Name

vlan

Comment

Netflow Profile

None

Config

IPv4

IPv6

Advanced

Assign Interface To

VLAN

Vlan Object

Virtual Router

default

Security Zone

Trust-L3

OK

Cancel

Switch to the IPv4 Tab

Click add and enter the IP address 192.168.1.254/24

VLAN Interface ?

Interface Name

Comment

Netflow Profile None ▼

Config | **IPv4** | IPv6 | Advanced

Type ☒ Static ☐ DHCP Client

☐ IP

☒ 192.168.1.254

+ Add - Delete ↑ Move Up ↓ Move Down

IP address/netmask. Ex. 192.168.2.254/24

OK

Cancel

Configure the DHCP Server

Go to Network > DHCP > DHCP server and click add

Edit to match the configuration shown below

DHCP Server ?

Interface vlan ▼

Mode enabled ▼

Lease | Options

☐ Ping IP when allocating new IP

Lease ☐ Unlimited ☒ Timeout

1 Days 0 Hours 0 Minutes

☐ IP POOLS ^

☒ 192.168.1.2-192.168.1.254

+ Add - Delete

RESERVED ADDRESS	MAC ADDRESS	DESCRIPTION
192.168.1.20	xx:xx:xx:xx:xx:xx	(Optional MAC Address)

+ Add - Delete

OK

Cancel

DHCP Server

Interface

vlan

Mode

enabled

Lease

Options

Inheritance Source

ethernet1/1

Check inheritance source status

Gateway

192.168.1.254

Subnet Mask

255.255.255.0

Primary DNS

inherited

Secondary DNS

inherited

Primary WINS

inherited

Secondary WINS

inherited

Primary NIS

inherited

Secondary NIS

inherited

Primary NTP

inherited

Secondary NTP

inherited

POP3 Server

inherited

SMTP Server

inherited

DNS Suffix

None

Custom DHCP options

	NAME	CODE	TYPE	VALUE
--	------	------	------	-------

+

 Add

-

 Delete

↑

 Move Up

↓

 Move Down

OK

Cancel

Define a Security Profile Group

Go to Objects > Security Profile Groups and Click Add

Configure the Security profile as Wanted

Security Profile Group

Name

Internet

Antivirus Profile

default

Anti-Spyware Profile

strict

Vulnerability Protection Profile

strict

URL Filtering Profile

default

File Blocking Profile

None

Data Filtering Profile

None

WildFire Analysis Profile

None

OK

Cancel

Configure Outbound Security Policy
Go to Policies > Security and Click Add
Enter a name and description

Security Policy Rule

General

Source

Destination

Application

Service/URL Category

Actions

Name

Internet Outgoing

Rule Type

universal (default)

Description

All Traffic to the Internet

Tags

Group Rules By Tag

None

Audit Comment

Audit Comment Archive

OK

Cancel

Add the Source Zone

Security Policy Rule

General

Source

Destination

Application

Service/URL Category

Actions

Any

Any

any

any

SOURCE ZONE

Trust-L3

+

-

Delete

Any

Any

any

any

SOURCE ADDRESS

+

-

Delete

Any

Any

any

any

SOURCE USER

+

-

Delete

Any

Any

any

any

SOURCE DEVICE

+

-

Delete

Negate

OK

Cancel

and the Destination Zone

Security Policy Rule

General

Source

Destination

Application

Service/URL Category

Actions

select

Any

any

DESTINATION ZONE

Untrust-L3

+

-

Delete

Any

Any

any

DESTINATION ADDRESS

+

-

Delete

Any

Any

any

DESTINATION DEVICE

+

-

Delete

Negate

OK

Cancel

Specify the Action as Allow

Security Policy Rule ⓘ

General | Source | Destination | Application | Service/URL Category | **Actions**

Action Setting

Action: **Allow** ▼

☐ Send ICMP Unreachable

Profile Setting

Profile Type: **Group** ▼

Group Profile: **Internet** ▼

Log Setting

☐ Log at Session Start

☒ Log at Session End

Log Forwarding: **None** ▼

Other Settings

Schedule: **None** ▼

QoS Marking: **None** ▼

☐ Disable Server Response Inspection

OK Cancel

Configure the Outbound NAT Policy

Go to Policies > NAT and Click Add

Configure a name and check IPv4 for NAT Type

NAT Policy Rule ⓘ

General | **Original Packet** | Translated Packet

Name: **Internet Outgoing**

Description:

Tags: ▼

Group Rules By Tag: **None** ▼

NAT Type: **ipv4** ▼

Audit Comment:

[Audit Comment Archive](#)

OK Cancel

Switch to Original Packet and Specify the Source Zone, Destination Zone, and Destination Interface

NAT Policy Rule ⓘ

General | **Original Packet** | Translated Packet

☐ Any

☐ SOURCE ZONE ^

☒ Trust-L3

+ Add - Delete

Destination Zone: **Untrust-L3** ▼

Destination Interface: **ethernet1/1** ▼

Service: **any** ▼

☒ Any

☐ SOURCE ADDRESS ^

+ Add - Delete

☒ Any

☐ DESTINATION ADDRESS ^

+ Add - Delete

OK Cancel

On the Translated Packet Tab Configure:

Translation Type: Dynamic IP and Port
Address Type: Interface Address
Interface: Ethernet1/1

NAT Policy Rule

General

Original Packet

Translated Packet

Source Address Translation

Translation Type

Dynamic IP And Port

Address Type

Interface Address

Interface

ethernet1/1

IP Address

None

Destination Address Translation

Translation Type

None

OK

Cancel

Configure the Management IP
Go to Device > Settings > Mangement
Copy the Configurations Below

Management Interface Settings

IP Type

☒ Static ☐ DHCP Client

IP Address

192.168.1.1

Netmask

255.255.255.0

Default Gateway

192.168.1.254

IPv6 Address/Prefix Length

Default IPv6 Gateway

Speed

auto-negotiate

MTU

1500

Administrative Management Services

☐ HTTP ☒ HTTPS

☐ Telnet ☒ SSH

Network Services

☐ HTTP OCSP ☒ Ping

☐ SNMP ☐ User-ID

☐ User-ID Syslog Listener-SSL ☐ User-ID Syslog Listener-UDP

☐

PERMITTED IP ADDRESSES

DESCRIPTION

+ Add

- Delete

OK

Cancel

Configure the DNS for Management
Go to Device > Setup > Services
Enter DNS Server IPs (Googles are 8.8.8.8 and 8.8.4.4)

Services

Services | NTP

Update Server

updates.paloaltonetworks.com

☒ Verify Update Server Identity

DNS Settings

DNS

☒ Servers
☐ DNS Proxy Object

Primary DNS Server

8.8.8.8

Secondary DNS Server

8.8.4.4

Minimum FQDN Refresh Time (sec)

30

FQDN Stale Entry Timeout (min)

1440

Proxy Server

Server

Port

[1 - 65535]

User

Password

Confirm Password

☐ Use proxy to send logs to Cortex Data Lake

OK

Cancel

Commit the Changes

Commit

Doing a commit will overwrite the running configuration with the commit scope.

☒ Commit All Changes
☐ Commit Changes Made By: 1 admin

COMMIT SCOPE	LOCATION TYPE	OBJECT TYPE	ENTITIES	ADMINS
device-and-network	Device and Network Configuration			

Preview Changes

Change Summary

Validate Commit

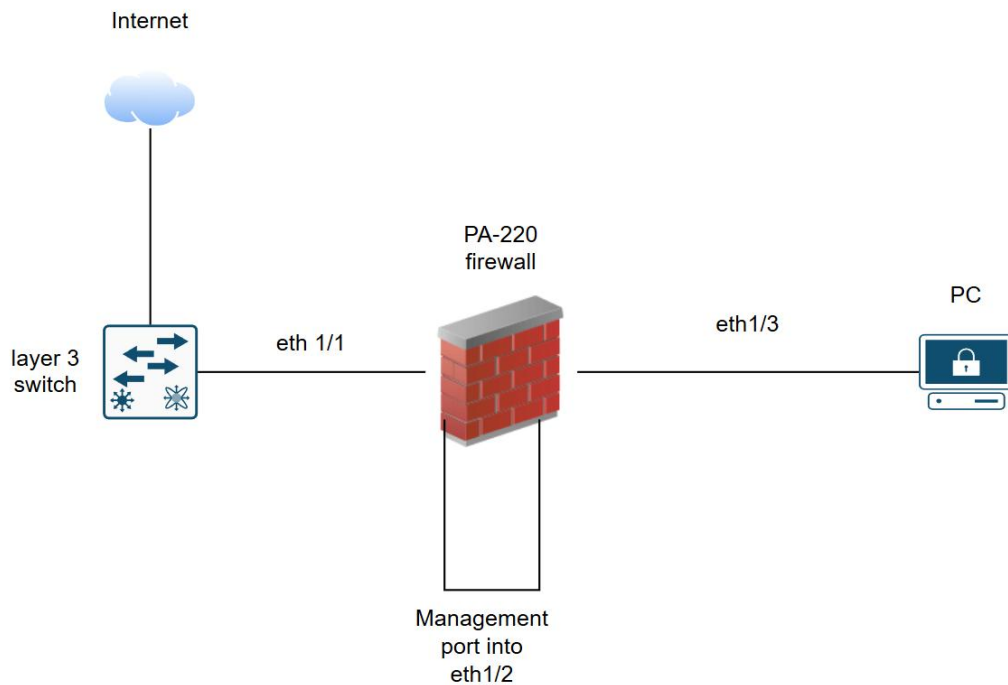
Note: This shows all the changes in login admin's accessible domain.

Description

Commit

Cancel

Network Diagram



Configuration

admin@PA-220> show config running

```
config {
  mgt-config {
    users {
      admin {
        phash $5$ncsoawar$a1lSJVF8dcteC7lLeG43sIFlEKjYIObfQnagfQA0z13;
        permissions {
          role-based {
            superuser yes;
          }
        }
      }
    }
  }
  password-complexity {
    enabled yes;
    minimum-length 8;
  }
}
```



```
shared {
  application;
  application-group;
  service;
  service-group;
  botnet {
    configuration {
      http {
        dynamic-dns {
          enabled yes;
          threshold 5;
        }
        malware-sites {
          enabled yes;
          threshold 5;
        }
        recent-domains {
          enabled yes;
          threshold 5;
        }
        ip-domains {
          enabled yes;
          threshold 10;
        }
        executables-from-unknown-sites {
          enabled yes;
          threshold 5;
        }
      }
      other-applications {
        irc yes;
      }
      unknown-applications {
        unknown-tcp {
          destinations-per-hour 10;
          sessions-per-hour 10;
          session-length {
            maximum-bytes 100;
          }
        }
      }
    }
  }
}
```

```
        minimum-bytes 50;
    }
}
unknown-udp {
    destinations-per-hour 10;
    sessions-per-hour 10;
    session-length {
        maximum-bytes 100;
        minimum-bytes 50;
    }
}
}
}
report {
    topn 100;
    scheduled yes;
}
}
content-preview {
    application;
    application-type {
        category {
            networking {
                subcategory {
                    remote-access;
                    proxy;
                    encrypted-tunnel;
                    routing;
                    infrastructure;
                    ip-protocol;
                    it-infrastructure;
                    it-management;
                }
            }
        }
    }
    collaboration {
        subcategory {
            email;
            instant-messaging;
```

```
social-networking;
internet-conferencing;
voip-video;
web-posting;
social-business;
productivity-tools;
}
}
media {
  subcategory {
    photo-video;
    gaming;
    audio-streaming;
    audio;
    ar-vr;
  }
}
business-systems {
  subcategory {
    auth-service;
    database;
    erp-crm;
    general-business;
    management;
    office-programs;
    software-update;
    storage-backup;
    software-development;
    marketing;
    coin-mining;
    medical;
    ics-protocols;
    vertical-industry;
    hr;
    recruiting;
    sales;
    commerce;
    security;
```

```
customer-service;
analytics;
digital-advertising;
design;
supply-chain-logistics;
artificial-intelligence;
cad-plm;
governance-risk-compliance;
iot-management;
b2b-marketplace-platforms;
content-management;
data-privacy;
education;
finance;
cryptocurrency;
}
}
general-internet {
  subcategory {
    file-sharing;
    internet-utility;
    hosting;
  }
}
saas {
  subcategory {
    vertical-industry;
    marketing;
    hr;
    erp;
    it-infrastructure;
    sales;
    development;
    commerce;
    security;
    customer-service;
    it-management;
    content-management;
```

collaboration-productivity;
office;
analytics;
digital-advertising;
design;
supply-chain-logistics;
hosting;
artificial-intelligence;
cad-plm;
governance-risk-compliance;
iot-management;
b2b-marketplace-platforms;
ar-vr;
data-privacy;
email;
erp-crm;
file-sharing;
management;
storage-backup;
internet-conferencing;
remote-access;
general-business;
internet-utility;
office-programs;
voip-video;
software-development;
social-business;
web-posting;
social-networking;
photo-video;
infrastructure;
instant-messaging;
marketing;
database;
auth-service;
general-business;
audio-streaming;
}

```
    }
    unknown;
  }
  technology {
    network-protocol;
    client-server;
    peer-to-peer;
    browser-based;
  }
}
}
}
devices {
  localhost.localdomain {
    network {
      interface {
        ethernet {
          ethernet1/1 {
            layer3 {
              ndp-proxy {
                enabled no;
              }
              sdwan-link-settings {
                upstream-nat {
                  enable no;
                  static-ip;
                }
                enable no;
              }
              lldp {
                enable no;
              }
              dhcp-client;
            }
          }
          ethernet1/2 {
            layer2 {
              lldp {
```

```
        enable no;
    }
}
ethernet1/3 {
    layer2 {
        lldp {
            enable no;
        }
    }
}
ethernet1/4 {
    layer2 {
        lldp {
            enable no;
        }
    }
}
loopback {
    units;
}
vlan {
    ipv6 {
        neighbor-discovery {
            router-advertisement {
                enable no;
            }
        }
    }
}
ndp-proxy {
    enabled no;
}
adjust-tcp-mss {
    enable no;
}
ip {
    192.168.1.254;
```

```

    }
}
tunnel {
    units;
}
}
vlan {
    "Vlan Object" {
        virtual-interface {
            interface vlan;
        }
        interface [ ethernet1/2 ethernet1/3 ethernet1/4];
    }
}
virtual-wire;
profiles {
    monitor-profile {
        default {
            interval 3;
            threshold 5;
            action wait-recover;
        }
    }
}
ike {
    crypto-profiles {
        ike-crypto-profiles {
            default {
                encryption [ aes-128-cbc 3des];
                hash sha1;
                dh-group group2;
                lifetime {
                    hours 8;
                }
            }
        }
        Suite-B-GCM-128 {
            encryption aes-128-cbc;
            hash sha256;
        }
    }
}

```



```
dh-group group19;
lifetime {
    hours 8;
}
}
Suite-B-GCM-256 {
    encryption aes-256-cbc;
    hash sha384;
    dh-group group20;
    lifetime {
        hours 8;
    }
}
}
ipsec-crypto-profiles {
    default {
        esp {
            encryption [ aes-128-cbc 3des];
            authentication sha1;
        }
        dh-group group2;
        lifetime {
            hours 1;
        }
    }
}
Suite-B-GCM-128 {
    esp {
        encryption aes-128-gcm;
        authentication none;
    }
    dh-group group19;
    lifetime {
        hours 1;
    }
}
Suite-B-GCM-256 {
    esp {
        encryption aes-256-gcm;
```



```

        priority low;
    }
    class7 {
        priority low;
    }
    class8 {
        priority low;
    }
    }
    }
    }
    }
    }
}
virtual-router {
    default {
        protocol {
            bgp {
                enable no;
                dampening-profile {
                    default {
                        cutoff 1.25;
                        reuse 0.5;
                        max-hold-time 900;
                        decay-half-life-reachable 300;
                        decay-half-life-unreachable 900;
                        enable yes;
                    }
                }
            }
            routing-options {
                graceful-restart {
                    enable yes;
                }
            }
        }
        rip {
            enable no;
        }
    }
}

```

```
ospf {
  enable no;
}
ospfv3 {
  enable no;
}
}
interface;
ecmp {
  algorithm {
    ip-modulo;
  }
}
}
default-vr {
  ecmp {
    algorithm {
      ip-modulo;
    }
  }
}
protocol {
  bgp {
    routing-options {
      graceful-restart {
        enable yes;
      }
    }
    enable no;
  }
  rip {
    enable no;
  }
  ospf {
    enable no;
  }
  ospfv3 {
    enable no;
  }
}
```

```
    }  
    interface [ vlan ethernet1/1];  
  }  
}  
dhcp {  
  interface {  
    vlan {  
      server {  
        option {  
          dns {  
            primary inherited;  
            secondary inherited;  
          }  
          wins {  
            primary inherited;  
            secondary inherited;  
          }  
          nis {  
            primary inherited;  
            secondary inherited;  
          }  
          ntp {  
            primary inherited;  
            secondary inherited;  
          }  
          lease {  
            timeout 1440;  
          }  
          pop3-server inherited;  
          smtp-server inherited;  
          gateway 192.168.1.254;  
          subnet-mask 255.255.255.0;  
          inheritance {  
            source ethernet1/1;  
          }  
        }  
      }  
    }  
    ip-pool 192.168.1.2-192.168.1.252;  
    mode enabled;
```

```
    }  
  }  
}  
}  
}  
deviceconfig {  
  system {  
    ip-address 192.168.1.1;  
    netmask 255.255.255.0;  
    update-server updates.paloaltonetworks.com;  
    update-schedule {  
      threats {  
        recurring {  
          weekly {  
            day-of-week wednesday;  
            at 01:02;  
            action download-only;  
          }  
        }  
      }  
    }  
  }  
  timezone US/Pacific;  
  service {  
    disable-telnet yes;  
    disable-http yes;  
  }  
  hostname PA-220;  
  device-telemetry {  
    device-health-performance yes;  
    product-usage yes;  
    threat-prevention yes;  
    region Americas;  
  }  
  type {  
    static;  
  }  
  default-gateway 192.168.1.254;  
  dns-setting {
```

```
servers {
  primary 8.8.8.8;
  secondary 8.8.4.4;
}
}
setting {
  config {
    rematch yes;
  }
  management {
    hostname-type-in-syslog FQDN;
  }
}
vsys {
  vsys1 {
    application;
    application-group;
    zone {
      trust {
        network {
          virtual-wire;
        }
      }
      untrust {
        network {
          virtual-wire;
        }
      }
      Untrust-L3 {
        network {
          layer3 ethernet1/1;
        }
      }
      Trust-L3 {
        network {
          layer3 vlan;
```

```

    }
  }
  Trust-L2 {
    network {
      layer2 [ ethernet1/2 ethernet1/3 ethernet1/4];
    }
  }
}
service;
service-group;
schedule;
rulebase {
  security {
    rules {
      rule1 72acc7ac-c7b4-4704-9ffa-7fc75cae80d9 {
        from trust;
        to untrust;
        source any;
        destination any;
        service any;
        application any;
        action allow;
        log-end yes;
      }
      "Internet Outgoing" 1144eee5-d331-4606-8333-31ded65f2210 {
        profile-setting {
          group Internet;
        }
        to Untrust-L3;
        from Trust-L3;
        source any;
        destination any;
        source-user any;
        category any;
        application any;
        service application-default;
        source-hip any;
        destination-hip any;
      }
    }
  }
}

```



```

        action allow;
        description "All Traffic to the Internet";
    }
}
}
nat {
    rules {
        "Internet Outgoing" 6c46de0f-b540-4cec-9e1c-7adefc699cf7 {
            to Untrust-L3;
            from Trust-L3;
            source any;
            destination any;
            service any;
            to-interface ethernet1/1;
            source-translation {
                dynamic-ip-and-port {
                    interface-address {
                        interface ethernet1/1;
                    }
                }
            }
        }
    }
}
import {
    network {
        interface [ ethernet1/1 ethernet1/2 ethernet1/3 ethernet1/4 vlan];
    }
}
address {
    "Outside Interface IP address" {
        ip-netmask 22.22.22.22;
        description "IP address for the Internet pointing Towards ISP";
    }
    "static route to isp next hop" {
        ip-netmask 0.0.0.0;
    }
}

```

